ModelIt!

Teacher's Guide

Can We Cure Cancer?

Modeling Cell Cycle Regulation, Oncogene Activation, and Immunotherapy Strategies

Quick Reference

Grade Level	Grade 12L3
NGSS Standard	HS-LS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: HS-LS1-4

Use a model to illustrate the role of cellular division (mitosis) and differentiation in producing and maintaining complex organisms.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a multi-component model to simulate the interaction between oncogenic mutations, tumor suppression, immune surveillance, and the tumor microenvironment, testing how different mutation combinations and treatment strategies affect cancer outcomes.
Disciplinary Core Idea	**LS1.B Growth and Development of Organisms** — Students model how disruption of normal cell division control mechanisms — through oncogene activation, tumor suppressor loss, and checkpoint failure — leads to uncontrolled proliferation and how the immune system provides an additional layer of growth regulation.
Crosscutting Concept	**Cause and Effect** — Students trace the causal chain from individual mutations through multiple cellular defense failures to tumor formation and immune escape, identifying how each cause-effect link creates a potential intervention target for cancer therapy.

CAST Testing Alignment

MULTIPLE CHOICE:

A cell acquires a KRAS oncogene mutation but retains functional TP53 tumor suppressor and intact cell cycle checkpoints. Based on the cancer model, what is the most likely outcome and why?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why checkpoint inhibitor immunotherapy is most effective in tumors with high PD-L1 expression and high tumor mutational burden, and least effective in tumors with low PD-L1 expression and an immunosuppressive microenvironment. Reference at least three model components and their relationships.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Cancer is not one disease but hundreds — each characterized by a unique combination of genetic mutations, immune evasion strategies, and microenvironment interactions. Yet all cancers share a common origin: normal cells that accumulate mutations allowing them to escape the body's multi-layered growth control systems. Understanding these control systems and how they fail is the foundation of modern cancer biology and the key to developing more effective treatments.

The Multi-Hit Model of Cancer

In 1971, Alfred Knudson proposed that cancer requires multiple genetic hits — mutations that disable successive layers of cellular growth control. We now know that most solid tumors carry 4-7 driver mutations in specific gene categories: oncogenes (growth accelerators), tumor suppressors (growth brakes), DNA repair genes (error correctors), and apoptosis regulators (self-destruct signals). The reason cancer incidence increases exponentially with age is that each additional year provides more cell divisions for random mutations to accumulate. A 70-year-old has undergone roughly 10 trillion more cell divisions than a 20-year-old, providing vastly more opportunities for the critical combination of mutations to occur in a single cell lineage.

Immune Surveillance and Escape

The immune system continuously patrols the body for abnormal cells. Cytotoxic T cells can recognize and kill cells displaying mutant proteins (neoantigens) on their surface. Natural killer cells detect cells that have lost normal surface markers. This immune surveillance destroys the vast majority of potentially cancerous cells before they can form tumors. However, successful cancers evolve immune evasion strategies: downregulating surface antigens to become invisible, expressing checkpoint proteins like PD-L1 to deactivate T cells, secreting immunosuppressive molecules, and recruiting regulatory T cells that suppress the immune response locally. The tumor microenvironment becomes an immunological dead zone.

The Immunotherapy Revolution

Traditional cancer treatments — surgery, chemotherapy, radiation — attack cancer cells directly but also damage normal tissue. Immunotherapy takes a fundamentally different approach: it empowers the patient's own immune system to attack the cancer. Checkpoint inhibitors (anti-PD-1, anti-PD-L1, anti-CTLA-4) block the molecular brakes that tumors use to suppress immune attack. CAR-T cell therapy genetically engineers a patient's own T cells to recognize and kill cancer cells. Cancer vaccines train the immune system to recognize tumor-specific antigens. Since 2011, immunotherapy has achieved lasting remissions in melanoma, lung cancer, kidney cancer, and blood cancers that were previously untreatable, earning James Allison and Tasuku Honjo the 2018 Nobel Prize in Medicine.

The Challenge of Solid Tumors

While immunotherapy has revolutionized treatment of some cancers, solid tumors remain the frontier challenge. Unlike blood cancers, solid tumors build physical barriers (dense fibrous stroma), create immunosuppressive microenvironments, restrict T cell infiltration, and develop heterogeneous subclone populations where different cells have different mutations and vulnerabilities. Combination therapies that simultaneously attack multiple vulnerabilities — checkpoint inhibitors plus targeted therapy plus anti-angiogenic agents — show the most promise. The future of oncology is personalized medicine: sequencing each patient's tumor to identify the specific combination of mutations, immune markers, and microenvironment characteristics, then designing a tailored multi-agent treatment protocol.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten key components of the cancer system: mutation rate, carcinogen exposure, inherited risk, oncogene activation, tumor suppressor function, checkpoint integrity, immune surveillance, microenvironment immunosuppression, angiogenesis, and treatment response.

E **Establish Relationships**

Students map the multi-hit pathway from initial mutation through progressive defense failure to malignancy, and the parallel immune surveillance pathway showing how tumors evolve evasion strategies.

V **Visualize & Model**

Students build a ten-component computational model showing the competing dynamics between cellular defense mechanisms and oncogenic progression, with the tumor microenvironment as a critical moderating variable.

E **Evaluate Outcomes**

Students run multi-hit progression, immune escape, and immunotherapy intervention scenarios to identify the conditions under which cancer develops, evades immunity, and responds to treatment.

R **Revise & Extend**

Students add metastasis potential, epigenetic dysregulation, or drug resistance evolution to model more realistic cancer biology and understand why treatment failures occur.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Right now, inside your body, cells are dividing billions of times a day. Some of those divisions produce errors — mutations. Your body has an extraordinary defense system that catches and destroys almost all of them. Cancer happens when that defense system fails. Today you will understand exactly how it fails and how we are engineering ways to fight back."*

Watch For

Show cover image. Ask: how many of you know someone who has been affected by cancer? Acknowledge the personal connections. Emphasize that understanding cancer biology is the path to better treatments.

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling the most complex disease humans face. Cancer is not one disease but hundreds, each with a unique combination of mutations. You will discover why cancer requires multiple failures, how the immune system fights back, and how immunotherapy is changing everything."*

Watch For

Have students read objectives. Pre-teach 'oncogene' and 'tumor suppressor' with the car analogy: oncogenes are stuck accelerators, tumor suppressors are the brakes. Cancer requires both the accelerator stuck AND the brakes failed.

Slide 3: Big Question

3 min

Say: *"Your immune system kills potentially cancerous cells every day. The fact that you do not have cancer right now is proof that your defense system is working. The question is: what happens when cancer evolves faster than your immune system can adapt?"*

Watch For

Quick-write: Students list everything they know about how cancer develops and how it is treated. Share out and organize into 'what causes cancer' and 'how we fight cancer' categories on the board.

Slide 4: LEVER Framework

2 min

Say: *"We will Label ten components of the cancer defense and attack system, Establish the multi-hit pathway and immune surveillance relationships, Visualize the model, Evaluate cancer progression and treatment scenarios, and*

Refine by adding metastasis and drug resistance."

Watch For

Preview each LEVER step. Emphasize that this model will explain not just how cancer develops but why treatments work (or don't) and what the future of cancer treatment looks like.

Slide 5: Activity 1: Components

8 min

Say: *"Three of these components represent risk factors you cannot fully control — mutation rate, carcinogen exposure, and inherited risk. Seven represent the body's defense systems and the tumor's counter-strategies. Sort them and identify which components are on your side and which are on the tumor's side."*

Watch For

Students sort external versus internal components. Discuss: some internal components protect you (tumor suppressors, checkpoints, immune surveillance) while others help the tumor (oncogene activation, microenvironment immunosuppression, angiogenesis). This framing helps students understand cancer as a battle.

Slide 6: Activity 2: Connections

8 min

Say: *"Trace the multi-hit pathway: first an oncogene activates, then a tumor suppressor fails, then checkpoints are compromised, then the immune system is evaded. At each step, the remaining defenses compensate — until they can't. Where is the tipping point?"*

Watch For

Students draw arrows showing how sequential mutations progressively compromise defenses. Identify the critical relationship between Tumor Microenvironment Immunosuppression and Immune Surveillance Effectiveness — this is where immunotherapy intervenes.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run three scenarios: watch cancer develop step by step through multi-hit progression, observe the immune system lose the battle against immune escape, and then unleash immunotherapy to see if we can turn the tide. Record what happens at each step."*

Watch For

Students run all three scenarios with detailed observations. Key moment: when they apply checkpoint inhibitor therapy and see Immune Surveillance reactivate against the tumor. Discuss: why does this work so dramatically in some patients and not others?

Slide 8: Discoveries

5 min

Say: *"Your model reveals four critical truths: cancer needs multiple mutations, the microenvironment is as important as the mutations, immunotherapy works by releasing existing immune power, and resistance evolves through natural selection. These insights are transforming cancer treatment right now."*

Watch For

Class discussion of discoveries. Ask: how does understanding cancer as an evolutionary process change how you think about treatment design? Why might a combination of therapies be better than any single therapy?

Slide 9: STEM Challenge

12 min

Say: *"You are the oncology team. Your patient has stage III lung cancer with a specific genomic profile. Use your model data to design a personalized treatment plan that targets this tumor's specific vulnerabilities. Every choice must be justified by model evidence."*

Watch For

Groups design treatment protocols based on the patient profile. Each team must specify drug combinations, treatment sequence, monitoring biomarkers, and contingency plans for resistance. Present and evaluate: which protocol is most likely to achieve lasting remission?

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Mutation Rate, Carcinogen Exposure, Inherited Genetic Risk
- **INTERNAL:** Oncogene Activation Level, Tumor Suppressor Function, Cell Cycle Checkpoint Integrity, Immune Surveillance Effectiveness, Tumor Microenvironment Immunosuppression, Angiogenesis Rate, Treatment Response Score

Relationships

- **Mutation Rate to Oncogene Activation Level:** POSITIVE (+) — Higher mutation rates increase the probability that growth-promoting proto-oncogenes will acquire activating mutations. Each cell division is a chance for a replication error to hit a critical oncogene.
- **Carcinogen Exposure to Mutation Rate:** POSITIVE (+) — Carcinogens directly damage DNA or promote cell proliferation that increases replication error opportunities. Tobacco smoke alone contains over 70 known carcinogens that cause specific mutation patterns in lung tissue.
- **Mutation Rate to Tumor Suppressor Function:** NEGATIVE (-) — Higher mutation rates increase the probability of inactivating mutations in tumor suppressor genes. Since both copies must be lost for complete inactivation, higher mutation rates accelerate the two-hit process.
- **Tumor Suppressor Function to Cell Cycle Checkpoint Integrity:** POSITIVE (+) — Key tumor suppressors like TP53 and RB1 are essential components of cell cycle checkpoints. When these genes are functional, checkpoints catch damaged cells; when they are lost, damaged cells divide unchecked.
- **Immune Surveillance Effectiveness to Treatment Response Score:** POSITIVE (+) — Stronger immune surveillance increases treatment response because many cancer therapies — especially immunotherapy — depend on immune cells to deliver the final killing blow against tumor cells.
- **Tumor Microenvironment Immunosuppression to Immune Surveillance Effectiveness:** NEGATIVE (-) — As the tumor microenvironment becomes more immunosuppressive through PD-L1 expression, regulatory T cell recruitment, and immunosuppressive cytokine secretion, immune surveillance is progressively disabled in the tumor locality.
- **Angiogenesis Rate to Tumor Microenvironment Immunosuppression:** POSITIVE (+) — New blood vessels supply the tumor with nutrients and also deliver immunosuppressive factors. Paradoxically, the abnormal blood vessels in tumors create a hypoxic, acidic environment that further suppresses immune cell function.
- **Oncogene Activation Level to Angiogenesis Rate:** POSITIVE (+) — Activated oncogenes like RAS promote the production of vascular endothelial growth factor (VEGF) that stimulates new blood vessel growth into the tumor, enabling continued growth beyond the 1-2mm diffusion limit.

Key Discoveries

- Cancer requires multiple mutations in specific combinations — the model reveals that a single oncogene activation or single tumor suppressor loss is usually insufficient for malignancy because the remaining defense layers (checkpoints, immune surveillance) compensate, explaining why cancer is rare before age 50
- The tumor microenvironment is as important as the tumor itself — a tumor that cannot suppress local immune responses or stimulate blood vessel growth is eliminated by the immune system or starved of nutrients, regardless of how many oncogenic mutations it carries
- Immunotherapy works by reactivating existing immune capabilities rather than adding new ones — checkpoint inhibitors do not create new immune cells but release the brakes on T cells that already recognize the tumor, which explains why immunotherapy is most effective in patients with strong but suppressed immune responses

- Treatment resistance is an evolutionary process — just as bacteria evolve antibiotic resistance, cancer cells with mutations that confer resistance to chemotherapy or immunotherapy survive and proliferate, making combination therapies that attack multiple vulnerabilities simultaneously more effective than single-agent approaches

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Cancer is caused by a single mutation"	Cancer requires the accumulation of 4-7 driver mutations in specific gene categories — oncogenes, tumor suppressors, DNA repair genes, and apoptosis regulators — in a single cell lineage. A single mutation is almost always insufficient because the body's redundant defense systems compensate. This is why inherited single-gene mutations like BRCA1 increase cancer risk but do not guarantee cancer — they provide a head start by eliminating one defense layer, but additional somatic mutations are still required.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The immune system has nothing to do with cancer"	The immune system is constantly detecting and destroying potentially cancerous cells — a process called immune surveillance. People with weakened immune systems (organ transplant recipients on immunosuppressive drugs, HIV/AIDS patients) have dramatically higher cancer rates, proving that immune surveillance is an active cancer defense. Immunotherapy drugs that enhance immune function have achieved lasting remissions in previously incurable cancers. The tumor's ability to evade immune detection is now recognized as one of the hallmarks of cancer.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Cancer is always a death sentence"	While some cancers remain very difficult to treat, five-year survival rates have improved dramatically. The overall five-year survival rate for all cancers combined has increased from 50% in the 1970s to over 68% today. Some cancers that were uniformly fatal — like certain melanomas and blood cancers — now have 50-90% response rates with immunotherapy. Early detection, targeted therapy, and combination approaches continue to improve outcomes. The model shows that treatment effectiveness depends on the specific combination of tumor characteristics, which is why personalized medicine is the future of oncology.